

Ashok Kumar Chittibomma

chashokkumar192@gmail.com

☎ 650-797-0810

Professional Summary

- Over 6+ years of IT experience in **Data Modeling, Data Engineering, Designing, Data architecture, and data analysis.**
- Experience building distributed high-performance systems using **Spark** and **Scala**.
- Experienced in **Spark Core, Spark RDD, Pair RDD, and Spark Deployment** Architectures.
- Experience in creating data lakes using **Spark** for downstream applications.
- Hands-on experience in the **Hadoop ecosystem**, including **Spark, Kafka, HBase, Scala, Impala, Sqoop, Oozie, Flume, Storm, and other big data** technologies.
- Proficient in building **real-time and batch data pipelines** using **Kafka, Spark Streaming, and Flume** for continuous data ingestion and analysis.
- Experience in optimizing **MapReduce** programs using combiners, partitioners, and custom counters for delivering the best results.
- Experience in writing scripts using Python scripting, **Python API, PySpark API, and Spark API** for data analysis.
- Experience developing **Airflow workflows** for scheduling and orchestrating the **ETL process**.
- Experience with **Informatica (ETL Tool)** for Data Extraction, Transformation, and Loading.
- Experienced in various reporting objects in **Tableau**, including Facts, Attributes, Joins, Sorting, Hierarchies, Transformations, Filters, Calculated Fields, Sets, Groups, and Parameters.
- Experience in **Machine Learning** and **Data Mining** with large datasets of structured and unstructured data, including **Data Acquisition, Data Validation, and Predictive Modeling**.
- Experienced in implementing various **Big Data Engineering, Cloud Data Engineering, Data Warehousing, Data Mart, Data Visualization, Reporting** and **Data Quality**, .
- Experience in **data transformation, data mapping** from source to target **database schema**, and **data cleansing procedures**.
- Experience in migrating SQL databases to **Azure Data Lake, Azure Data Lake Analytics, Azure SQL Database, Databricks, and Azure SQL Data Warehouse**, including controlling and granting database access.
- Hands-on experience working with **Amazon Web Services (AWS)** using **Elastic Map Reduce (EMR), Redshift, and EC2** for data processing.
- Hands-on experience in Star Schema Modeling, **Snow-Flake** Modeling, **FACT** and **Dimensions Tables**, and **Physical and Logical Data Modeling** using **Erwin**.
- Experienced in developing data pipelines using **Kafka** to store data into **HDFS**.
- Experience developing **Scala** applications for loading/streaming data into **NoSQL databases (MongoDB)** and **HDFS**.
- Leveraged expertise in **database design and implementation**, including **data modeling, SQL, distributed computing, operating systems, cloud computing, storage technologies, memory management, and networking**, to build efficient and **scalable data solutions**

Technical Skills

Category	Skills
Databases	Snowflake, AWS RDS, Teradata, Oracle, MySQL, Microsoft SQL, PostgreSQL
NoSQL Databases	MongoDB, Hadoop HBase, Apache Cassandra
Programming Languages	Python, SQL, Scala, MATLAB
Cloud Technologies	Azure, AWS, GCP
Data Formats	CSV, JSON
Querying Languages	SQL, NoSQL, PostgreSQL, MySQL, Microsoft SQL
Integration Tools	Jenkins
Scalable Data Tools	Hadoop, Hive, Apache Spark, Pig, MapReduce, Sqoop
Operating Systems	Red Hat Linux, Unix, Windows, macOS
Reporting & Visualization	Tableau, Matplotlib

Professional Experience

Client: American Airlines, Dallas, TX

Feb 2024 - Present

Role: Data Engineer

Responsibilities:

- Involved in analysis, specification, design, and implementation and testing phases of **Software Development Life Cycle (SDLC)** and used **agile** methodology for developing application.
- Established and managed **AWS S3 buckets** to securely and efficiently store large datasets, ensuring both data integrity and easy access.
- Involved in writing **Spark** applications using **Scala** to perform various data cleansing, validation, transformation, and summarization activities according to the requirement.
- Designed and implemented **real-time data pipelines** using **Kafka and Spark Streaming** for large-scale data ingestion.
- Developed **Scala** based **Spark** applications for performing data cleansing, event enrichment, data aggregation, de-normalization and data preparation needed for machine learning and reporting teams to consume.
- Developed **MapReduce** programs for Data Analysis and worked on compression mechanisms to optimize **MapReduce** Jobs.
- Designed and implemented configurable data delivery pipeline for scheduled updates to customer facing data stores built with **Python**.
- Worked on developing **Pyspark** script to encrypting the raw data by using Hashing algorithms concepts on client specified columns.
- Built **ETL packages** in **SSIS** to perform full loads as well as incremental loads from OLTP to data warehouse using different data flow and control flow tasks and provide maintenance for existing jobs.
- Created **Power BI** reports and dashboards as per the business requirement using different data sources.
- Developed **Tableau** data visualization using **Cross tabs, Heat maps, Bar Charts, Gantt charts, Waterfall charts, Scatter Plots, Geographic Maps, Pie Charts and Donut Charts**.
- Developed and Configured **Kafka brokers** to pipeline server logs data into **spark** streaming.
- Used **Apache Kafka** to aggregate web log data from multiple servers and make them available in Downstream systems for analysis.
- **AWS S3 & Snowflake**: Managed secure data storage and deployed scalable, high-performance data warehousing on AWS and Snowflake.
- **DevOps & Automation**: Automated deployment and infrastructure management using **Jenkins, Chef, Maven**, and **AWS CloudFormation Templates**.
- Built and maintained robust **data pipelines** for real-time and batch processing using **AWS Glue, AWS Data Pipelines, GCP BigQuery**, and **Fivetran**.
- Optimized **AWS Redshift** and **AWS RDS** for efficient, high-performance data storage and transactional processing.
- Implemented **AWS ELB, CloudWatch**, and **SQS** for enhanced load balancing, real-time monitoring, and scalable messaging between microservices.
- Led the **migration** of legacy systems to AWS, improving system performance while minimizing downtime.
- Enhanced ETL workflows with **dbt, Apache Airflow**, and **Fivetran** to streamline data ingestion, transformation, and orchestration.
- Designed and implemented **disaster recovery strategies** and **cloud-based backup solutions** to ensure data protection and business continuity.
- Led the integration of **healthcare data** with **Epic EMR**, transforming complex clinical data into pipelines for better decision-making.
- Monitor and maintain **CI/CD pipelines** in **Azure** environment for data load from Lake to DBX and from DBX to SQL DW.
- Used **Airflow** for scheduling the **Hive, Spark, and MapReduce** jobs.
- Implemented **Apache Airflow** for authoring, scheduling and monitoring Data Pipelines.
- Design and implement secure data pipelines into a **Snowflake data warehouse** from **on-premises** and **cloud data sources**.
- Created **Hive** tables to store the processed results in a tabular format.
- Writing the script files for processing data and loading to **HDFS**.
- Implemented a Continuous Delivery pipeline with **Docker**, and **Git Hub**.
- Used **Jira** for bug tracking.
- Worked on designing, building, deploying and maintaining **Mongo DB**.
- Maintained and developed complex **SQL queries, stored procedures, views, functions** in **MS SQL server**.

- Involved in story-driven **agile** development methodology and actively participated in daily scrum meetings.
- Participated in the status meetings and status updating to the management team.
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Environment: JIRA 7.2.x, Confluence6.2, Remedy9.1, Maven 3.0.2, Artifactory 5.4.5, GITHUB 2.0., Ubuntu, CentOS, Linux, AWS ELB, AWS SQS, AWS S3, AWS Cloud Formation Templates, Snowflake, AWS RDS, AWS Cloud Watch, PowerShell 4.2.6, AWS, AWS Redshift, AWS S3, AWS Data Pipe Lines, AWS Glue, DevOps, jQuery, Tomcat, Apache, Jenkins 1.0.5, Python3.6.0, Ruby2.3, Chef12.7.2, **AWS Redshift, Snowflake, Apache Tomcat, AWS Glue, Jenkins, Python, Data Governance Tools** (for governance-related tasks), **Data Modeling Tools** like Erwin, **Data Cataloging Tools** (e.g., AWS Glue Data Catalog).

Client: Paychex, Rochester, NY

July 2023 - Jan 2024

Role: Data Engineer

Responsibilities:

- Engineered and upheld robust data solutions utilizing **Azure SQL and Azure SQL MI**, enhancing data storage and processing for optimal availability and performance.
- Crafted and implemented scalable **Azure Function App and Azure WebApp solutions** to streamline data workflows and user interactions.
- Leveraged **SQL Server 2017** for the design, implementation, and maintenance of databases, ensuring superior performance and dependability for crucial data operations.
- Created sophisticated data analytics and visualization tools using **Power BI and Tableau**, empowering stakeholders with data-driven decision-making capabilities.
- Enhanced continuous integration and delivery of data projects through the automation of data ingestion and transformation processes using **Azure DevOps**.
- Worked as **Data Engineer** to review business requirement and compose source to target data mapping documents..
- Developed **Spark** applications by using **Scala** and **Python** and implemented **Apache Spark** for data processing from various streaming sources.
- Developed **Spark** scripts by using **Scala** per the requirement to read/write **JSON** files.
- Developed monitoring and notification tools using **Python**. Wrote **Python** routines to log into the websites and fetch data for selected options. Used Collections in **Python** for manipulating and looping through different user defined objects.
- Developed **PySpark** script to merge static and dynamic files and cleanse the data.
- Developed **Map Reduce** program for parsing and loading into **HDFS**.
- Performed **Data Analysis, Data Migration, Data Cleansing, Transformation, Integration, Data Import, and Data Export** through **Python**.
- Developed interactive dashboards and reports using **Power BI** for day-to-day business decision making and strategic planning needs.
- Enhanced **ETL** packages performance by adjusting properties, restructure flows and tuned embedded queries by adjusting indexes using execution plan and data engine tuning advisor.
- Used **Tableau** Desktop to analyze and obtain insights into large data sets using groups, hierarchies, sorts, sets and filters.
- Extract Transform and Load data from Sources Systems to **Azure Data Storage services** using a combination of **Azure Data Factory, T-SQL, Spark SQL and U-SQL Azure Data Lake Analytics Data Ingestion** to one or more **Azure Services - (Azure Data Lake, Azure Storage, Azure SQL, Azure DW)** and processing the data in **In Azure Databricks**.
- Implement ad-hoc analysis solutions using **Azure Data Lake Analytics/Store, HDInsight**.
- Worked with **Data governance, Data quality, data lineage, Data architect** to design various models and processes.
- Created **Airflow** Scheduling scripts in **Python**.
- Performed Real time event processing of data from multiple servers in the organization using **Apache Storm** by integrating with **apache Kafka**.
- Implemented data ingestion and handling clusters in real time processing using **Kafka**.
- Handled importing data from various data sources, performed transformations using **Hive, Map Reduce**, and loaded data into **HDFS**.

- Design and implement secure data pipelines into a **Snowflake data warehouse** from **on-premises** and **cloud data sources**.
- Worked on **MongoDB** to load and retrieve data for real time processing using **Rest API**.
- Design and development of **SQL Server, T-SQL scripts** and functions for data import, data export, data conversion, and data cleansing.
- Involved in **Agile** methodologies, daily scrum meetings, spring planning.
- Actively participated and provided feedback in a constructive and insightful manner during weekly Iterative review meetings to track the progress for each iterative cycle and figure out the issues.

Environment: Azure Function App, Azure WebApp, Azure SQL, and Azure SQL MI, SSH, YAML, WebLogic, Python, Azure DevOps, Git, Maven, Jira, SQL Profiler, Database Engine Tuning Advisor, Jira, GIT Hub, SharePoint, : SQL Server 2017, Power BI, SFDC, SQL, T-SQL, Hive, Microsoft Business Intelligence Studio, Microsoft Visual Studio Data Tools, Hue, TableauSpark, Scala, Python, PySpark, MapReduce, ETL, Tableau, Power BI, Azure, Snowflake, StarSchema, Apache Kafka, Apache Airflow, Hive, HDFS, MongoDB, SQL, Agile and Windows.

Client: Kaiser Permanente, Denver, CO (Remote, India)

Sept 2019 - Dec 2022

Role: Data Engineer

Responsibilities:

- **Data Warehousing & Cloud Storage:** Developed and managed large-scale data warehousing solutions using **GCP BigQuery** and **Cloud SQL**, optimizing data storage, retrieval, and analytics for business intelligence.
- **ETL Pipeline Engineering:** Engineered high-performance **ETL pipelines** using **Matillion**, automating data transformation and integration from multiple sources into **GCP BigQuery**, ensuring scalability and data integrity.
- **Serverless Architecture:** Designed and deployed cost-efficient, serverless data pipelines leveraging **AWS Lambda** and **Azure Functions**, optimizing performance while reducing infrastructure costs.
- **Advanced SQL & Performance Tuning:** Crafted complex **SQL queries** and implemented **partitioning** and **clustering** in **GCP BigQuery**, driving improved performance, cost optimization, and faster data retrieval.
- **Data Modeling & Governance:** Led the design and implementation of **enterprise-wide canonical data models** in **GCP BigQuery** and **Cloud SQL**, ensuring alignment with data governance policies and enterprise architecture standards.
- **Machine Learning & Data Processing:** Leveraged **GCP Databricks** for extensive data processing and machine learning, accelerating data-driven decision-making and operational efficiency across the organization.
- **Real-Time Data Streaming:** Administered **Pub/Sub** for real-time data streaming, enabling an event-driven architecture that enhanced system scalability and responsiveness.
- **Cloud Security & Compliance:** Implemented strong data security protocols, including **VPC**, **VPN**, and **IAM**, ensuring secure data transfers and compliance with **GDPR** and **HIPAA** standards.
- **Data Quality & Consistency:** Ensured high data quality standards with **QualityStage**, maintaining accuracy and consistency in data used for critical business insights and decision-making.
- **Automation & Efficiency:** Automated routine data tasks and system operations using **Python** and shell scripts, reducing manual errors and boosting operational efficiency.
- **Cloud Cost Optimization:** Optimized cloud resource usage with **GCP CloudWatch**, monitoring performance and managing costs to ensure efficient cloud operations
- Developed **Hive** queries and **UDFS** to analyze/transform the data in **HDFS**.
- Performed Data cleaning and Preparation on **XML** files.
- Used **SQL** queries and other tools to perform data analysis and profiling.
- Involved in **Agile** methodologies, daily scrum meetings, spring planning.
- Actively participating in the code reviews, meetings and solving any technical issues.
- **Leadership & Mentorship:** Provided technical leadership and mentorship to junior data engineers, fostering a collaborative environment and driving continuous improvement across data engineering practices.

Environment: GCP Databricks, Service Data Transfer, python, shell scripts VPC Configuration, Data Catalog. VPN Google- Client, Pub Sub, SSIS, SSAS, SSRS, DATASTAGE, QUALITYSTAGE, MS-SQL, ORACLE, DB2, Federated Queries, IAM Security, GCP, GCP Big Query, Spark, Scala, ETL, Tableau, Python, PySpark, Snowflake, Apache Spark, Hive, AWS, XML, JSON, HDFS, SQL, Agile and Windows 10.

Client: Xoom Works, India

May 2018 - Aug 2019

Role: Data Engineer

Responsibilities:

- Involved in gathering the business requirements and interacted with business analysts to translate business requirements into technical specifications.
- Developed **Spark** application by using **Python (PySpark)** to transform data according to business rules.
- Involved in developing production ready **spark** application using **Spark RDD APIs, Data frames, Spark-SQL and Spark-Streaming API's**.
- Built models using **Python** and **PySpark** to predict probability of attendance for various campaigns and events.
- Developed **Spark-Streaming** applications to consume the data from **Kafka topics** and to insert the processed streams to **HBase**.
- Developed **Map Reduce** program for parsing and loading into **HDFS** information.
- Involved in building Data Integration, Workflow Solutions and **Extract, Transform, and Load (ETL)** solutions for data warehousing using **SQL Server Integration Service (SSIS)**.
- Develop user-friendly, informative **Tableau** dashboards based on the client requirements; publish the developed dashboards on to the production site after the testing of their usability on the test site.
- Performing **Sqoop** jobs to land data on **HDFS** and running validations.
- Creating **Hive** tables, loading and analyzing data using hive scripts. Implemented Partitioning, Dynamic Partitions, Buckets in **HIVE**.
- Working with **JSON, XML** file formats.
- Extensively involved in writing SQL queries (sub queries and join conditions) for building and testing ETL processes.
- Participated in the status meetings and status updating to the management team.

Environment: Spark, Scala, Python, PySpark, ETL, Tableau, HDFS, SSIS, XML, JSON, Agile and Win

Education

Master of Science in Computer Science
Southern University and A&M College

Jan 2023 – May 2024

Bachelor of Science in Mechanical Engineering
Sathyabama University

Jun 2014 - Apr 2018